Semantic-Guided 3D Gaussian Splatting For Sparse View Reconstruction And Segmentation

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Why Is Sparse-View 3DGS Hard?

Geometry Ambiguity

Sparse point cloud

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Unstable reconstruction

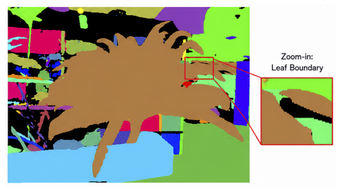
Novel-View Distortion

Distorted rendering

⇕

Reference view

Boundary Inconsistency



Predicted segmentation

⇕

Reference segmentation

Sparse-view 3DGS is underconstrained: limited observations produce ambiguous geometry, distorted novel views, and inconsistent object boundaries.

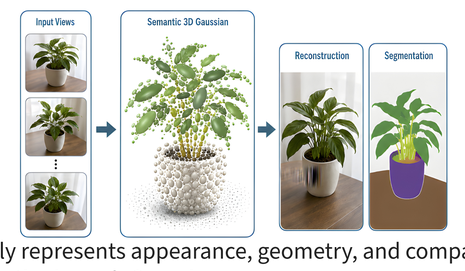
Core Insight

Semantics guide geometry ⇔ Geometry sharpens segmentation.

SegSplat turns semantic information into an active geometric prior for joint sparse-view reconstruction and object-level understanding.

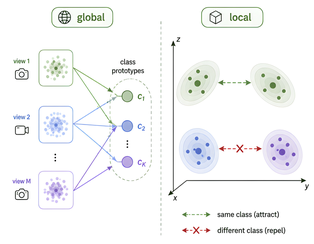
Key Contributions

Unified Semantic 3DGS



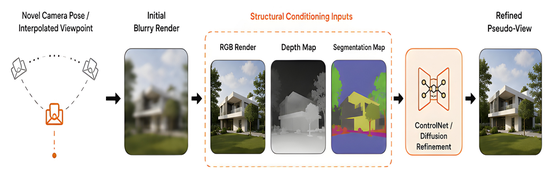
Jointly represents appearance, geometry, and compact semantics in each Gaussian.

Multi-Level Semantic Consistency



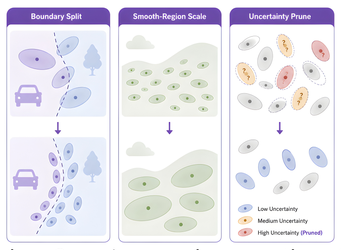
Combines DINO-distilled prototypes with 3D class regularization for coherent objects and sharp boundaries.

Pose-Aware View Augmentation



Uses depth- and segmentation-guided diffusion to supervise unseen viewpoints.

Semantics-Aware Density Control



Splits boundary Gaussians, scales smooth regions, and prunes semantically uncertain Gaussians.

Ablation Highlights

Complementary Semantic Supervision

+0.36 PSNR	+0.96 mIoU	+1.78 bIoU

Full supervision compared with CE alone.

Objective	PSNR↑	mIoU↑	bIoU↑
CE	22.48	95.55	71.72
CE + KD	21.84	95.17	71.11
CE + DPCL	22.48	96.19	72.10
CE + DPCL + KD	**22.84**	**96.51**	**73.50**

DPCL and foundation-model distillation contribute complementary semantic cues.

Local 3D Class Regularization

+0.53 PSNR	+3.21 mIoU	+3.81 bIoU

Gain from enforcing local 3D class consistency.

Setting	PSNR↑	mIoU↑	bIoU↑
Without	22.31	93.30	69.69
With regularization	**22.84**	**96.51**	**73.50**

Local consistency improves object coherence, boundaries, and reconstruction quality.

Geometry-Conditioned Pseudo-Views

+0.93 PSNR	−0.020 LPIPS	+2.11 bIoU

Improvement from supervising unseen viewpoints.

Setting	PSNR↑	LPIPS↓	bIoU↑
Without pseudo-views	21.91	0.162	71.39
With pseudo-views	**22.84**	**0.142**	**73.50**

Geometry-consistent pseudo-views improve unseen-view fidelity and semantic boundaries.

Semantics-Aware Density Control

+1.67 PSNR	+1.35 mIoU	+5.47 bIoU

Full density control compared with standard densification.

B	BG	P	PSNR↑	mIoU↑	bIoU↑
0	0	0	21.17	95.16	68.03
1	0	0	22.63	96.21	73.23
0	1	0	22.18	96.05	72.51
0	0	1	22.39	96.33	71.70
1	1	0	22.61	96.02	71.53
1	**1**	**1**	**22.84**	**96.51**	**73.50**

B: boundary split **BG**: background scale **P**: uncertainty prune.

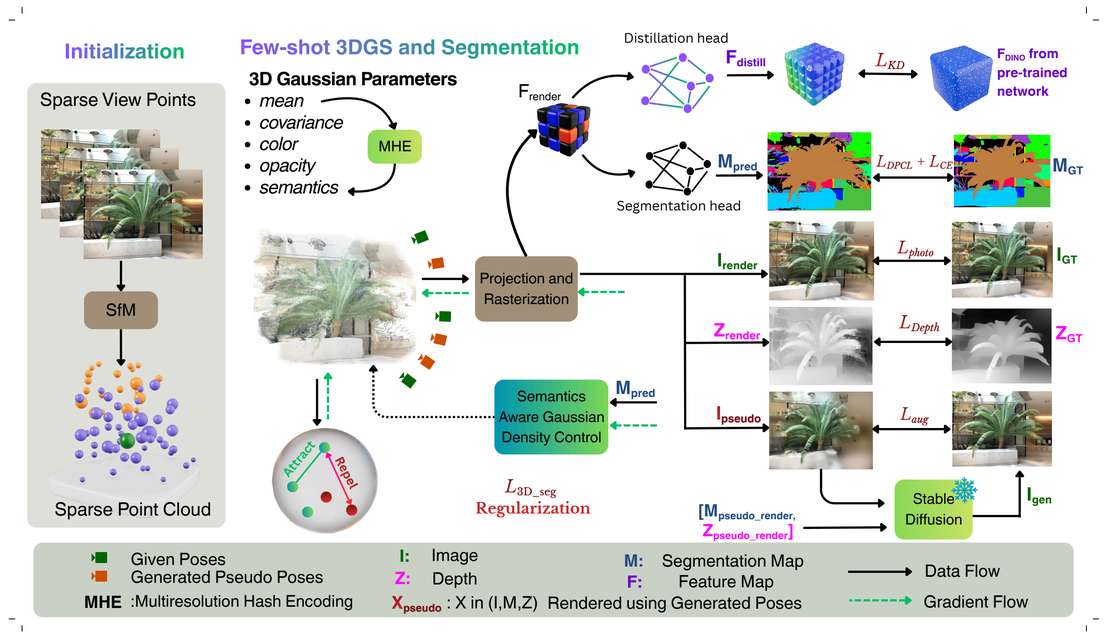
All ablations use Tanks & Temples. Each component contributes a distinct improvement to the joint representation.

SCAN ME

Selected References

[1] Kerbl et al., *3D Gaussian Splatting for Real-Time Radiance Field Rendering*, TOG 2023.
[2] Zhu et al., *FSGS: Real-Time Few-Shot View Synthesis Using Gaussian Splatting*, ECCV 2024.
[3] Ye et al., *Gaussian Grouping: Segment and Edit Anything in 3D Scenes*, ECCV 2024.

Methodology: SegSplat Pipeline



Sparse Initialization
SfM point cloud → 3D Gaussians

→

Joint Prediction
Render RGB, depth, and semantics

→

Novel-View Supervision
Diffusion-refined pseudo-views

Semantic optimization: teacher distillation + segmentation supervision + 3D class regularization + semantics-aware densification and pruning.

Joint Optimization

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{photo}} + \mathcal{L}_{\text{2D-seg}} + \mathcal{L}_{\text{3D-seg}} + \mathcal{L}_{\text{aug}} + \mathcal{L}_{\text{depth}}$$

Jointly optimizes appearance, 2D and 3D semantics, augmented views, and geometric consistency.

Sparse-View Novel-View Synthesis

Dataset	Best Prior per Metric PSNR↑	SSIM↑	LPIPS↓	SegSplat PSNR↑	SSIM↑	LPIPS↓
T&T (3 views)	22.45	0.772	0.156	**22.84**	**0.834**	**0.142**
LLFF (3 views)	**20.77**	**0.725**	**0.218**	20.69	0.710	0.220
Mip-NeRF 360 (24 views)	23.89	0.747	0.217	**24.16**	**0.753**	**0.215**

SegSplat is best across all NVS metrics on T&T and Mip-NeRF 360, while remaining competitive on LLFF.

Qualitative comparison on Mip-NeRF 360, LLFF, and T&T. SegSplat preserves fine details, thin structures, and global geometry while reducing blur and ghosting.

Segmentation and Object-Level Consistency

Sparse-View Segmentation

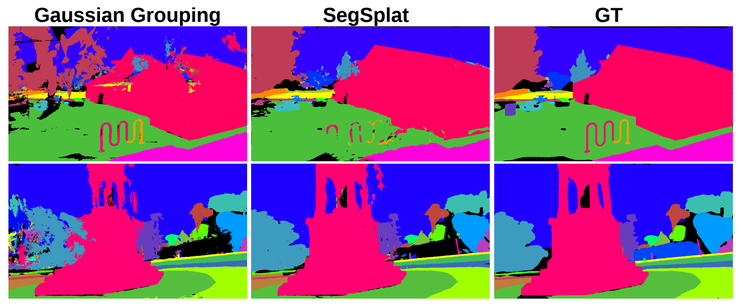
Dataset	Gaussian Grouping mIoU↑	Gaussian Grouping bIoU↑	SegSplat mIoU↑	SegSplat bIoU↑	△mIoU
LLFF	84.79	66.56	**88.18**	**67.59**	+3.39
Mip-NeRF 360	81.71	**50.68**	**82.14**	50.66	+0.43
T&T	87.29	56.95	**96.51**	**73.50**	**+9.22**

Largest gain on T&T: **+9.22 mIoU** and **+16.55 bIoU**.

NVOS Benchmark

Method	mIoU↑	mAcc↑
NVOS	70.1	92.0
Gaussian Grouping	85.6	97.3
FlashSplat	91.8	**98.6**
SAGA	90.9	98.3
COB-GS	92.1	**98.6**
SegSplat	**93.0**	**98.6**

Best mIoU despite sparse-view training.



SegSplat produces sharper object boundaries and more coherent semantic regions than Gaussian Grouping under sparse-view supervision.

Conclusion

SegSplat achieves 22.84 PSNR on T&T and 24.16 PSNR on Mip-NeRF360, improves T&T segmentation by +9.22 mIoU and +16.55 bIoU, and reaches 93.0 mIoU on NVOS.

SegSplat delivers high-fidelity reconstruction and consistent object-level understanding from sparse input views.

https://github.com/smeenapadnekar/SegSplat